

IA Mathematics for Natural Sciences A: Examples Sheet 3 - Vector area, polar coordinates, and complex numbers

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Questions marked with a (*) are difficult and should not be attempted at the expense of the other questions.

Vector area

1. (a) Define the *vector area* $\mathbf{A}$ of a connected, flat surface that has area A and unit normal $\hat{\mathbf{n}}$. What is the convention that is normally used when choosing the direction of the normal?
(b) Determine the vector areas of the: (i) parallelogram spanned by $\mathbf{a}, \mathbf{b}$; (ii) triangle spanned by $\mathbf{a}, \mathbf{b}$.
(c) The velocity $\mathbf{u} = (1, 0, 1)^T$ of a fluid is uniform everywhere. Compute the volume passing the triangle spanned by $\mathbf{a} = (0, 0, 1)^T, \mathbf{b} = (0, 1, 0)^T$ per unit time. [*This is called the velocity flux.*]
2. (a) Define the vector area $\mathbf{A}$ of a connected, non-flat surface composed piecewise of flat surfaces of areas $A_1, \dots, A_k$ with unit normals $\hat{\mathbf{n}}_1, \dots, \hat{\mathbf{n}}_k$ respectively. What do we require about the normals?
(b) Find the vector area $\mathbf{S}$ of the surface S made up of the two triangles OAB, BCO , where $A = (1, 0, 0), B = (1, 1, 1), C = (0, 2, 0)$. Is $|\mathbf{S}|$ equal to the surface area of S ?
3. Give a very general outline of how the concept of vector area might be extended to *curved* surfaces S ; hence, explain why we expect the vector area of a *closed* surface to be $\mathbf{0}$. Thus, find the vector areas of:
(a) a square-based pyramid with vertices $(0, 0, 0), (2, 0, 0), (2, 2, 0), (0, 2, 0), (1, 1, 1)$, with base removed;
(b) the unit hemisphere $x^2 + y^2 + z^2 = 1, z \geq 0$.
4. (a) Let $\mathbf{S}$ be the vector area of the surface S . Prove that the area of the projection (or 'shadow') of the surface S onto the plane with unit normal $\hat{\mathbf{m}}$ is $|\mathbf{S} \cdot \hat{\mathbf{m}}|$.
(b) Using the result from part (a), compute the area *and* the vector area of the projection of the square with vertices $(0, 0, 0), (2, 0, 0), (2, 2, 0)$ and $(0, 2, 0)$ onto a plane with normal $\mathbf{m} = (0, -1, 1)$.

Polar coordinate systems

5. Draw convincing diagrams defining plane polar, cylindrical, and spherical coordinates. Hence, find the cylindrical polar and spherical polar coordinates of the point $(3, 4, 5)$.
6. (a) In 2D Cartesians, a circle is specified by $(x - 1)^2 + y^2 = 1$. Find its equation in polars.
(b) In 3D Cartesians, a sphere is specified by $(x - 1)^2 + y^2 + z^2 = 1$. Find its equation in sphericals.
7. Let $a > 0$ be a constant. Describe the following loci:
(a) (i) $\phi = a$; (ii) $r = \phi$, in plane polar coordinates;
(b) (i) $z = a$; (ii) $r = a$; (iii) $r = a$ and $z = \phi$, in cylindrical polar coordinates;
(c) (i) $\theta = a$; (ii) $\phi = a$; (iii) $r = a$; (iv) $r = \theta = a$, in spherical polar coordinates.
8. Consider a point with position vector $\hat{\mathbf{n}}$ on the unit sphere S centred on the origin.
(a) Show that $\hat{\mathbf{n}} = (\sin(\theta) \cos(\phi), \sin(\theta) \sin(\phi), \cos(\theta))^T$, where (r, θ, ϕ) are polar coordinates.
(b) Show that the vector area $d\mathbf{S}$ of a small patch near $\hat{\mathbf{n}}$, subtending a small angle $d\theta$ in the θ direction, and a small angle $d\phi$ in the ϕ direction, is given *approximately* by $d\mathbf{S} = \hat{\mathbf{n}} \sin(\theta) d\theta d\phi$.
(c) (*) Hence, use integration to verify that the vector area of the unit sphere is $\mathbf{0}$.

Complex number basics

9. By expanding in real and imaginary parts, show that for any complex numbers z, w , we have:

$$\operatorname{Re}(z) = \frac{z + z^*}{2}, \quad \operatorname{Im}(z) = \frac{z - z^*}{2i}, \quad (z + w)^* = z^* + w^*, \quad (zw)^* = z^*w^*, \quad \left(\frac{z}{w}\right)^* = \frac{z^*}{w^*}.$$

Verify these results when $z = 1 + i$ and $w = 2 - 5i$.

10. (a) Factorise, and hence solve, the following polynomial equations:

$$(i) z^2 + 1 = 0; \quad (ii) z^2 + (1 - i)z - i = 0; \quad (iii) z^3 - (4 + 2i)z^2 + (4 + 5i)z - (1 + 3i) = 0.$$

(b) Prove that the solutions to a polynomial equation with real coefficients come in complex conjugate pairs. Deduce that if such an equation has odd degree, it must have at least one real root.

(c) How does the result of part (b) relate to the equations in part (a)?

Modulus and argument of a complex number

11. (a) Define the *modulus* and (*principal*) *argument* of the complex number z . Find the modulus and (*principal*) argument of the complex numbers $z = 2 + i$ and $w = 3 + 4i$.

(b) By writing z, w in modulus-argument form, show that:

$$|zw| = |z||w|, \quad \arg(zw) = \arg(z) + \arg(w), \quad \left|\frac{z}{w}\right| = \frac{|z|}{|w|}, \quad \arg\left(\frac{z}{w}\right) = \arg(z) - \arg(w).$$

Hence give geometrical interpretations of complex number multiplication and division. Verify these results for the complex numbers z, w from part (a).

12. By considering the product of the complex numbers $1 + iA, 1 + iB$, derive a formula expressing the sum $\arctan(A) + \arctan(B)$ as a single arctangent.

13. Let $Q = zw^*$, where z, w are complex numbers.

(a) Give a geometrical interpretation of the real and imaginary parts of Q , if z, w are regarded as two-dimensional vectors. By considering a vector area, deduce that the area of the triangle formed by the points $0, z, w$ in the complex plane is $\frac{1}{2} |\operatorname{Im}(zw^*)|$.

(b) Show that Q is invariant if z, w are both rotated through the same angle, and relate your result to your previous geometrical results.